



# Shipment Sure



**Predicting On-Time  
Delivery Using Supplier  
Data**

# Project Overview



## Problem Statement

To develop an explainable machine learning system that predicts shipment delivery delays and provides actionable insights through an interactive dashboard.

## GOAL

To develop an AI-based system that predicts shipment delays and provides explainable, real-time insights through an interactive dashboard.



# Key Feature of this project



Machine Learning-Based Prediction



End-to-End ML Pipeline



Prediction Probability & Confidence Score



Explainable AI (SHAP Integration)



Interactive Dashboard (Streamlit UI)



Prediction History Tracking



# Work Flow



**User Input**

**Data Preprocessing**

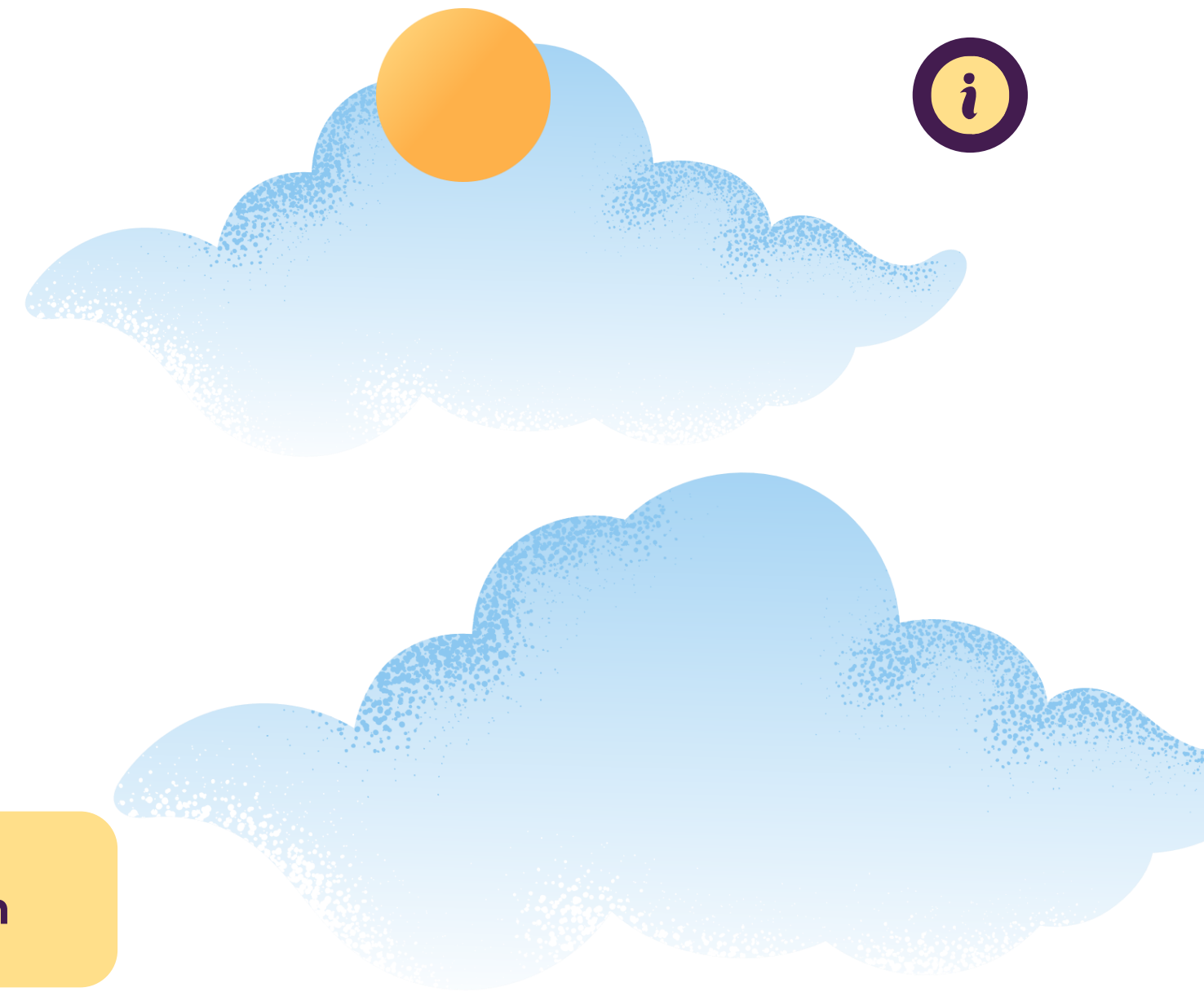
**Machine Learning  
Prediction**

**Probability &  
Confidence Calculation**

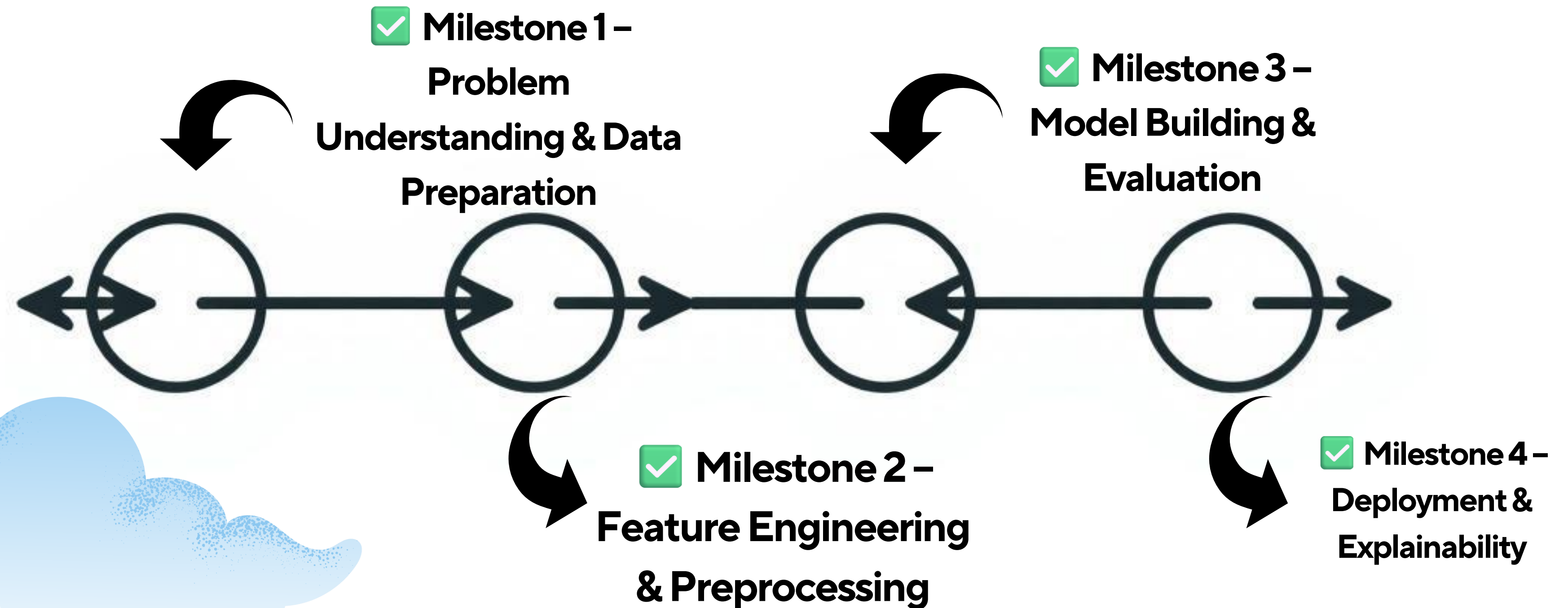
**Explainable AI (SHAP  
Analysis)**

**Result Visualization**

**Prediction History  
Tracking**



# MILESTONE OVERVIEW





# MILESTONE 1 OVERVIEW

- 1. Understood the problem of shipment delays in logistics**
- 2. Identified important factors that affect delivery (distance, supplier rating, weather, etc.)**
- 3. Collected the dataset related to shipment deliveries**
- 4. Cleaned the data by:**
  - **Removing missing or incorrect values**
  - **Converting data into proper format**
- 5. Prepared the data so that it can be used for machine learning**

 **Outcome:**

**Clean and ready dataset + clear understanding of the problem.**





# MILESTONE 2 OVERVIEW

- **Selected the most important features from the dataset**
- **Converted categorical values (like region, shipment mode) into numbers**
- **Scaled numerical values so all features are on the same level**
- **Built a preprocessing pipeline to handle all these steps automatically**

 **Outcome:**

**Well-structured data that machine learning models can understand**



# MILESTONE 3 OVERVIEW

- Trained different machine learning models to predict:

On-time delivery

Delayed delivery

- Tested the models using accuracy and other metrics
- Selected the best performing model
- Saved the trained model and preprocessing pipeline for future use

## Outcome:

- A trained and tested machine learning model ready for prediction.

**Best selected model is Logistic Regression with Accuracy 92.5%**

1. Accuracy

2. Precision And Recall

3. F1-Score

4. Confusion Matrix

5. ROC-AUC

6. GridSearch



# MILESTONE 4 OVERVIEW



- 1.Created a Streamlit web dashboard**
- 2.Allowed users to enter shipment details**
- 3.Displayed prediction results with confidence scores**
- 4.Added explainable AI (SHAP) to show why a shipment may be delayed**
- 5.Stored prediction history for analysis**

 **Outcome:**

**A complete, interactive, and explainable shipment delay prediction system.**



# Thank You



**Any Question feel free to  
ask**